Conditional probability

Lecture and worked examples

Maths Explorer

Reference course

In a weighted tree, the probabilities leaving the same node sum to 1. We multiply the probabilities along a path: $P(A \cap B) = P(A)P_A(B)$. We add the probabilities of mutually exclusive paths leading to the event we are looking for.

Reference course

Do not confuse $P_A(B)$ and $P_B(A)$. The condition comes after the word “given” and is written as a subscript in this notation.

Reference course

40% of customers buy online. Among them, 12% return their order; among the others, 5% return it. Then $P(R) = 0.40 \times 0.12 + 0.60 \times 0.05 = 0.078$. The overall return rate is therefore 7.8%.

Conditioning

- Define a conditional probability.
- Move from counts to probabilities.
- Interpret.

Conditioning

In a survey, 180 customers out of 600 use a premium card and 72 of these 180 customers buy every week. Calculate $P(P)$, $P(P \cap H)$ and $P_P(H)$, then interpret.

Conditioning

For $P(A) > 0$, $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. This is the probability of B within the population restricted to A.

Conditioning

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Conditioning

One-hour plan: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.

Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

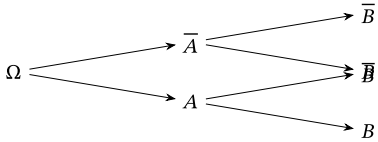
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Weighted trees

- Construct and complete a tree.
- Read a conditional probability.
- Check the sums.

Weighted trees

35% of orders are express. Among them, 4% are late; among standard orders, 9% are late. Construct and complete the tree.



Weighted trees

In a tree, the branches leaving a node sum to 1. The second-level probabilities are conditional.

Weighted trees

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Weighted trees

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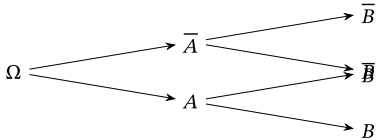
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Product of probabilities along a path

- Calculate an intersection.
- Combine paths.
- Translate an event.

Product of probabilities along a path

Using the tree from session 40, calculate the probability of an order being express and late, then the total probability of a delay. Interpret both results.



Product of probabilities along a path

We multiply along a path: $P(A \cap B) = P(A)P_A(B)$. We add the mutually exclusive paths that realise the event.

Product of probabilities along a path

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Product of probabilities along a path

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Independence

- Test independence.
- Compare probabilities.
- Justify a conclusion.

Independence

In a population, $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.12$. Test independence in two ways. Then change $P(A \cap B)$ to 0.15 and start again.

Independence

A and B are independent if $P(A \cap B) = P(A)P(B)$, equivalently if $P_A(B) = P(B)$ when $P(A) > 0$.

Independence

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Independence

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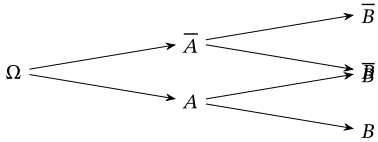
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Partition and total probability

- Recognise a partition.
- Apply total probability.
- Organise the calculation.

Partition and total probability

A factory has three lines A, B, C producing 20%, 50% and 30% of the items respectively. Their defect rates are 1%, 2% and 4%. Calculate the overall defect rate.



Partition and total probability

If (A_i) is a partition, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. Each term corresponds to a path leading to B.

Partition and total probability

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Partition and total probability

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Problems in context

- Choose a tree or a table.
- Solve a problem in context.
- Interpret.

Problems in context

Turn a table of counts (customers/subscription/cancellation) into a tree, then answer a question in each direction: probability of cancelling given the subscription; probability of having the subscription given the cancellation.

Problems in context

Choosing between a tree and a table: the tree highlights the order of conditioning; the table suits counts and reversed readings well.

Problems in context

Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Problems in context

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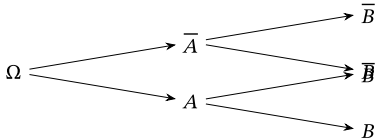
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Mixed practice

- Calculate a reverse probability.
- Chain total probability and conditioning.
- Check the result.

Mixed practice

In the situation of session 43, given that an item is defective, calculate the probability that it comes from line C. Explain why this result is not 30%.



Mixed practice

For a reverse probability, first calculate the total probability of the conditioning event, then use the definition of conditional probability.

Mixed practice

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Mixed practice

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Review and assessment

- Use all the methods.
- Write up with the notation.
- Remedy mistakes.

Review and assessment

Complete a four-station circuit: complete a tree; calculate an intersection; apply the total probability formula; test independence. Each group leaves a written justification for the next group.

Review and assessment

Always name the events, complete all the branches, write the formula before calculating and interpret the final probability.

Review and assessment

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